

Exp 3 wireless technology

RFID and Keypad Password-Based Door Lock System

Aim:

To design and implement a secure door lock system using RFID and keypad-based authentication with Arduino Mega. This project enhances security by combining two-factor authentication (RFID and password).

Theory:

An RFID and Keypad Password-Based Door Lock System is a modern security solution that uses Radio Frequency Identification (RFID) and a keypad to control access to a secure area.

1. RFID (Radio Frequency Identification):

- RFID is a wireless communication technology that uses radio waves to identify and authenticate objects or individuals.
- An RFID card or tag contains a unique identification number stored in its microchip.
- When the card is brought close to the RFID reader, the reader detects the ID and sends it to the Arduino for verification.

2. Keypad Authentication:

- A keypad is used to input a numeric or alphanumeric password.
- If the entered password matches the stored password, the system proceeds to unlock the door.
- This adds an additional layer of security to the system.

3. Servo Motor and Lock Mechanism:

- A servo motor is used to control the locking mechanism of the door.
- When the correct card and password are detected, the servo rotates to unlock the door.
- If the authentication fails, the door remains locked.

4. LCD Display:

- The LCD display provides real-time feedback to the user.
 - It shows messages like "Access Granted" or "Access Denied" based on the authentication result.
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Components:

- Arduino Mega – Acts as the central controller.
 - RFID Module (RC522) – Reads the RFID card/tag ID.
 - Keypad (4x4) – Takes user input for password authentication.
 - Servo Motor – Controls the door lock mechanism.
 - I2C LCD Display Module (16x2) – Displays system status.
 - Jumper Wires – For connecting components.
 - Breadboard – For easy wiring and component placement.
 - RFID Cards and Tags – Used for authentication.
 - Lock Mechanism – Physical locking system.
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Working:

1. System Initialization:

- The Arduino initializes and displays "System Ready" on the LCD.
- The servo motor sets the door to a locked position.

2. Reading RFID Card:

- When the user places an RFID card near the reader, the reader reads the unique ID.
- If the ID matches the stored value, the system proceeds to the next step.

3. Password Verification:

- The user is prompted to enter the password using the keypad.
- If the password is correct, the servo unlocks the door and displays "Access Granted."
- If the password is incorrect, "Access Denied" is shown on the LCD.

4. Locking the Door:

- After a few seconds, the servo motor returns to the locked position.
 - The system resets and waits for the next input.
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Code:

```
#include <Servo.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

Servo myServo;
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```

String password = "1234"; // Set the correct password
String input = "";

void setup() {
    lcd.begin();
    lcd.print("System Ready");
    myServo.attach(9);
    myServo.write(0); // Lock position
}

void loop() {
    char key = keypad.getKey();
    if (key) {
        input += key;
        lcd.setCursor(0, 1);
        lcd.print(input);

        if (input.length() == 4) {
            if (input == password) {
                lcd.clear();
                lcd.print("Access Granted");
                myServo.write(90); // Unlock position
                delay(3000);
                myServo.write(0); // Lock again
            } else {
                lcd.clear();
                lcd.print("Access Denied");
                delay(1000);
            }
            input = "";
            lcd.clear();
            lcd.print("System Ready");
        }
    }
}

```

Conclusion:

The RFID and Keypad Password-Based Door Lock System is a secure and efficient solution for access control. By combining RFID technology with password authentication, it provides

two-layer security. The system is user-friendly, cost-effective, and suitable for various security applications.